

«Pre-Conditions»

1. System state: Staff database is online.
2. 'alice' is authenticated and holds an accessLevel ≥ 5 .
3. 'bob' and 'charlie' exist in the system as active staff.

alice : WarehouseManager

staffId = 101
email = "alice@warehouse.com"
department = "Operations"
accessLevel = 5

manages

manages

manages

adminRole : Role

roleId = 1
roleName = "Admin"

bob : Picker

staffId = 102
email = "bob@warehouse.com"
pickZone = "Zone A"

charlie : Packer

staffId = 103
email = "charlie@warehouse.com"
packStation = "Station 1"

«Post-Conditions»

1. 'alice' is actively linked to the Admin role.
2. 'bob' is successfully allocated to Pick Zone A.
3. 'charlie' is successfully allocated to Pack Station 1.

«Pre-Conditions»

1. The "Building A" facility is registered and active in the system.
2. The Master Item Catalog contains the definition for SKU "BP-100".
3. Environmental monitoring systems for the refrigerated zone are online.

mainFacility : Warehouse

warehouseId = 1
location = "Building A"
capacity = 50000
companyRate = 4.5

coldZone : Zone

zoneId = "Z-Cold"
zoneType = "Refrigerated"
temperature = -4.0
humidity = 40.0

frozenMeatInv : Inventory

inventoryId = 55
quantity = 200
salesStock = 150
reorderLevel = 50

contains

beefPatty : Item

itemId = 998
name = "Beef Patty Box"
sku = "BP-100"
weight = 50.0
expiryDate = "2026-12-01"

«Post-Conditions»

1. 'coldZone' is successfully allocated within 'mainFacility' and tracking at -4.0°C.
2. 200 units of "Beef Patty Box" are physically secured in 'coldZone'.
3. 150 units are explicitly cleared and available for sales fulfillment.

«Pre-Conditions»

1. "TechCorp" is registered as an approved, active vendor in the system.
2. SupplierContract 'contract2026' (version 2) is valid, signed, and unexpired.
3. The Item catalog contains an active entry for SKU "BS-X1".

techCorp : Supplier

supplierId = 201
companyName = "TechCorp"
contactInfo = "sales@techcorp.com"

contract2026 : SupplierContract

contractId = 5001
version = 2
discountThreshold = 10.5

generates

po1044 : PurchaseOrder

poid = 1044
digitalSignature = "0xABC123"
estimatedArrivalDate = "2026-04-20"
status = Passed

orders

scanner : Item

itemId = 801
name = "Barcode Scanner"
sku = "BS-X1"

«Post-Conditions»

1. PurchaseOrder 'po1044' is successfully created and set to Passed status.
2. The order is digitally signed and legally bound to the terms of 'contract2026'.
3. The warehouse system forecasts incoming inventory of Barcode Scanners for 2026-04-20.

«Pre-Conditions»

1. Order 992 is verified, paid, and the required inventory is confirmed in stock.
2. The Picker 'bob' is clocked in, active, and available for a new task.
3. The warehouse routing engine is online to calculate the optimal path.

order992 : Order

orderId = 992
status = PROCESSING
totalAmount = 250.00

generates

pickList77 : PickList

pickListId = 77
optimizedRoute = "A1 -> B2 -> C1"
status = "Active"

assignedTo

bob : Picker

staffId = 102
pickZone = "Zone A"

worksIn

zoneA : Zone

zoneId = "Z-A"
zoneType = "Dry Goods"

«Post-Conditions»

1. 'pickList77' is successfully generated and bound to 'order992'.
2. 'bob' is exclusively assigned to 'pickList77' until the status changes.
3. The system expects Bob's physical location to begin in "Zone A" (Dry Goods).

«Pre-Conditions»

1. All physical items for Order 992 have been successfully picked and delivered to Station 1.
2. 'charlie' is logged in at Pack Station 1 and ready to process items.
3. The external carrier API (FedEx) is reachable to generate valid tracking numbers.

order992 : Order

orderId = 992
status = PACKING

packedBy

charlie : Packer

staffId = 103
packStation = "Station 1"

creates

shipment404 : Shipment

shipmentId = 404
status = "Ready"
trackingNumber = "1Z9999"
weight = 12.5

assignedTo

fedEx : Carrier

carrierId = 88
name = "FedEx Express"

«Post-Conditions»

1. 'shipment404' is fully manifested with a verified weight (12.5) and tracking number.
2. The physical package is formally staged on the dock for FedEx pickup.
3. The system is ready to transition 'order992' from PACKING to SHIPPED status.

«Pre-Conditions»
1. The safety threshold for 'Z-Cold' is configured strictly below 5.0°C.
2. 'tempSensor1' is active, calibrated, and physically wired to the monitoring network.
3. The warehouse emergency notification dispatch system is online.

coldZone : Zone
zoneId = "Z-Cold"
temperature = 5.0

tempSensor1 : IoTSensor
sensorId = 301
type = "Temperature"
lastReading = 5.0
nextMaint = "2026-06-01"

triggers

tempSpike : EmergencyEvent
eventId = 911
severity = "HIGH"
timestamp = "2026-04-14T13:40:00Z"
resolved = false

creates

alert99 : Alert
alertId = 99
message = "CRITICAL: Zone Z-Cold Temp above threshold!"
resolved = false

notifies

alice : WarehouseManager
staffId = 101

«Post-Conditions»
1. 'tempSpike' is formally logged in the system as a HIGH severity incident.
2. 'alert99' is successfully generated and dispatched to 'alice' (WarehouseManager).
3. The system locks both the event and alert state (resolved = false) pending manual review.

«Pre-Conditions»

1. The scheduled monthly audit for 2026-04-01 is initiated.
2. The physical stock count for 'widgetInv' is completed and verified as exactly 0.
3. The automated exception handling and replenishment system is active.

monthlyAudit : Audit

auditId = 701
date = "2026-04-01"
discrepancyRate = 0.02

checks

widgetInv : Inventory

inventoryId = 42
quantity = 0

triggers

missingWidget : BackOrder

backOrderId = 551
reason = "Audit Discrepancy"

creates

rushPO : PurchaseOrder

poid = 1045
status = EXPECTED

«Post-Conditions»

1. The system records a 2% discrepancy rate (0.02) for the audited sector.
2. BackOrder 'missingWidget' is formally lodged to track the unexpected shortage.
3. PurchaseOrder 'rushPO' is automatically generated and routed to the supplier to replenish the stock.